Honors Peer-graded Project

Summary

1. Introduction

Purpose of the Project

The main goal of this analysis is to uncover actionable insights from the New York City Airbnb Open Data. This dataset is fascinating because it captures a wide range of listings—covering diverse neighborhoods, price ranges, and lodging types—offering a rich foundation to explore how geographic, temporal, and host-level factors influence traveler choices, pricing, and occupancy. The final goal is to help investors to make the best choices to get an RBnB.

2. Context :

Airbnb has radically transformed the hospitality industry by empowering everyday homeowners to offer rental spaces to travelers worldwide. In a vibrant city like New York, which draws millions of visitors annually, understanding how hosts set prices, how guests select neighborhoods, and what influences occupancy rates can provide critical insights to maximise gain. By delving into this dataset, we can spot trends that matter not only to hosts aiming to optimize their listings but also to made the best choices for the rentability of the rental.

Description of the Dataset

This DataSet is a very popular because of it size and number of features. It require several steps to exploit data and learn Machine Learning and data analysis.

It contain 16 features :

Feature	Description
id	Unique id to identify Rbnb
name	Name of the rental
Host_id	Unique id to identify the host
Host_name	Name of the host (i.e. John, Lola..)
Neighbourhood_group	Location
neighbourhood	area
latitude	Coordinate
longitude	Coordinate
room_type	Category of the rented part (i.e. entire home, shared room, private room..)
price	Price of the rental in dollars
minimum_nights	Number of night required
number_of_reviews	Number of total reviews
last_review	Last review made on the house

reviews_per_month	Number of review per month made
calculated_host_listings_count	amount of listing per host
availability_365	How many days is the house avalaible for booking

3. Initial Plan for Data Exploration

We'll start by exploring the dataset to spot patterns, outliers, and any notable trends. Specifically, we want to understand how price varies across different neighborhoods, whether hosts with multiple listings have different review patterns, and if there's a relationship between room type and availability.

To achieve this, we'll:

- Generate basic descriptive statistics (mean, median, standard deviation) for features like price and number of reviews.
- Use histograms and box plots to see the distribution of price, availability, and minimum nights, and to detect outliers.
- Create scatter plots (e.g., price vs. number of reviews, or latitude/longitude to visualize geographic trends).
- Build a correlation heatmap to quickly identify potential relationships among numerical features.

All analyses will be done using Plotly for interactive visualizations in a Jupyter Notebook environment. This will allow for a clear, detailed, and dynamic exploration of the data.

4. Data Cleaning and Feature Engineering

Data Cleaning Steps

The first step is to delete duplicate row. In our case there was no duplicate ones.

```
# Check for duplicates and remove them
print("Number of rows before removing duplicates:", len(df))
df.drop_duplicates(inplace=True)
```

```
Number of rows before removing duplicates: 48895
Number of rows after removing duplicates: 48895
```

The second step is to identify missing values and decide a strategy to handle them.

Value name	Number of missing values	Strategy
name	16	Delete row (irreplaceable value, small size)
Host_name	21	Delete row (irreplaceable value, small size)
Last_review	10052	Don't modify
reviews_per_month	10052	Replace by 0 as they are no reviews

Then ensure all rows are in the good data format : price are strings :

```
# Convert 'price' to numeric safely
df = df.assign(price=pd.to_numeric(df['price'], errors='coerce'))
```

And done !

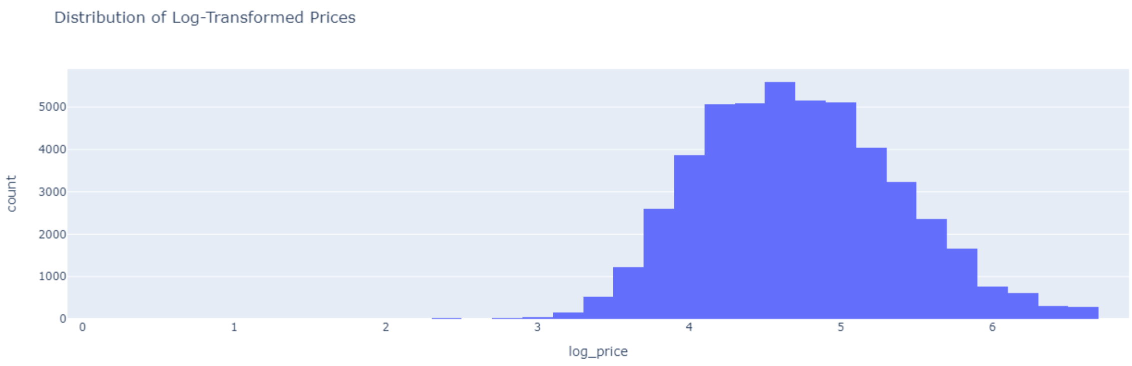
Last step, handle extreme values in prices - using IQR filtering removinf the 5% most exptreme values in our case – and ensure they're not negative. Remove extreme minimum nights (>30 nights) and houses available more than 365 days a year and modify to 365 if needed.

Feature Engineering

Here what i've done :

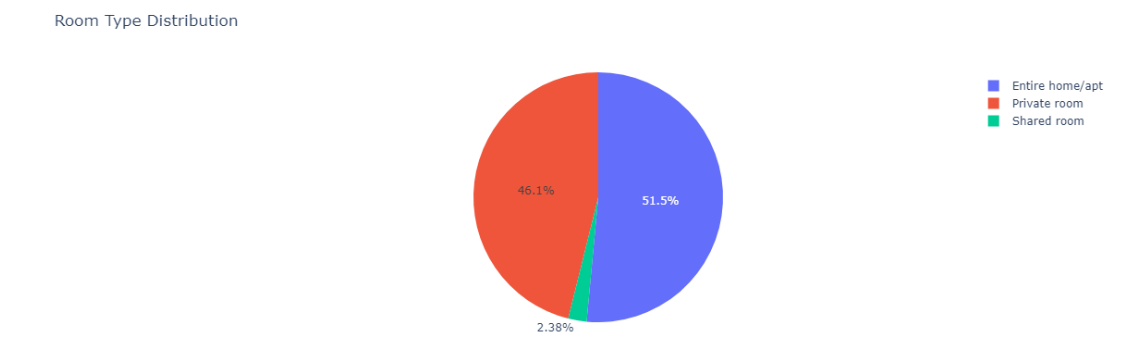
- **Day of Week for Reviews :** Helps analyze when users are more likely to leave reviews.
- **Price Binning :** Groups prices into meaningful ranges (Budget, Mid-range, etc.).
- **Normalization (MinMaxScaler) :** Ensures features like 'minimum_nights' and 'availability_365' are comparable.
- **Log Price Transformation :** Reduces skew in price distribution.
- **One-Hot Encoding :** Converts categorical features into machine-readable numerical format.
- **Estimated revenue :** let's suppose that it's alway booked, then estimated_revenue = price x availability (maximum possible revenue)
- **Occupancy rate :** based on the number of availability days on the year

Let’s visualise our data to see if it keep it consistency

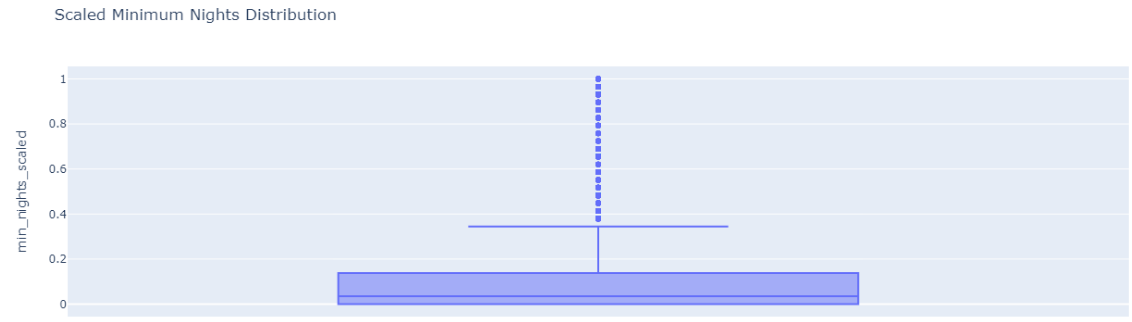


Price are excellent and describe a Gaussian like Curve

Prices are simplied to avoid a large amound of points. Here again it keep it consistency.



Room types are well converted to numeric values to be exploitable

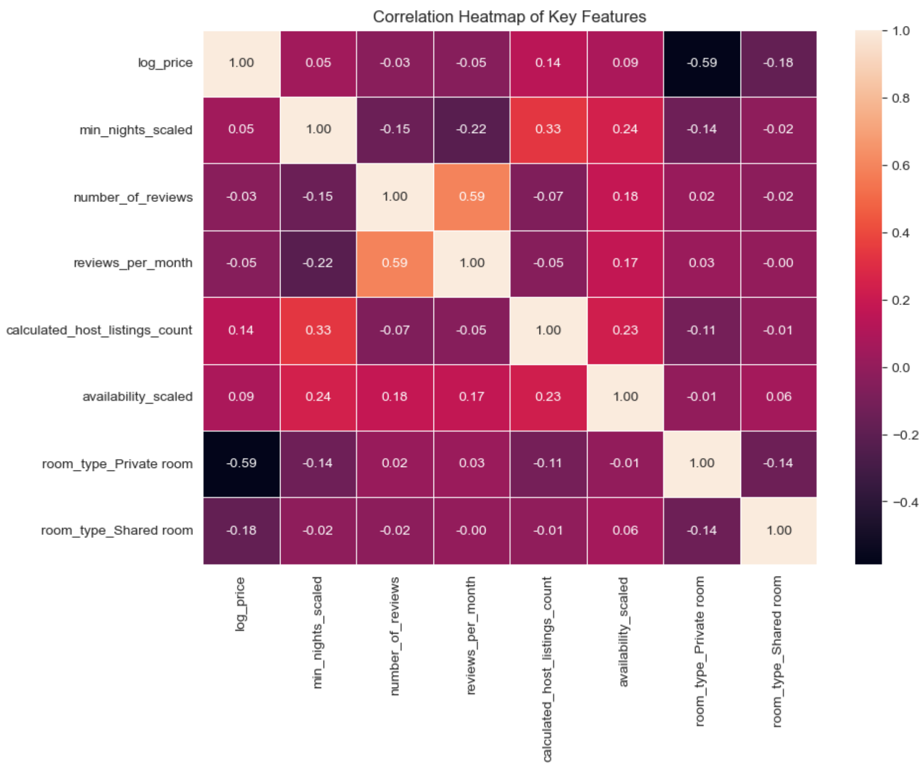


Finally Night distribution are logic, after extreme values are removed.

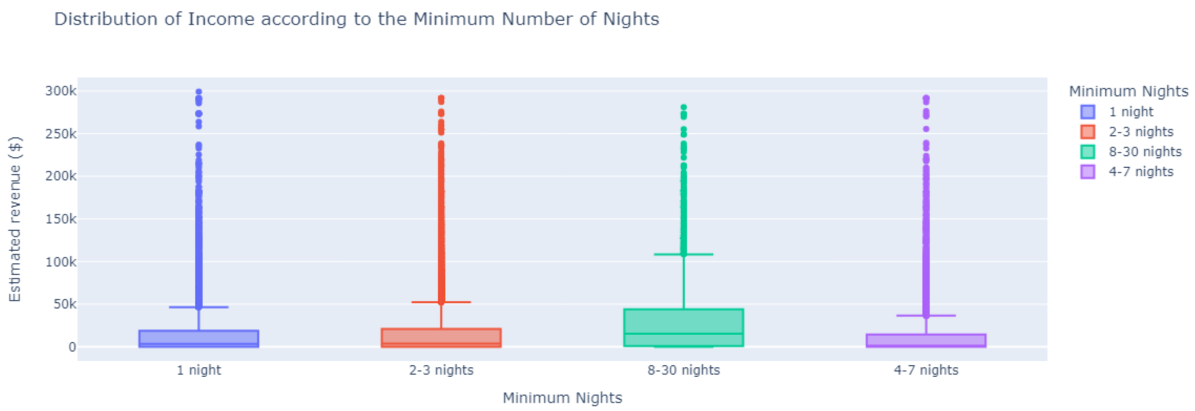
5. Data exploration

Exploratory Data Analysis (EDA) Summary

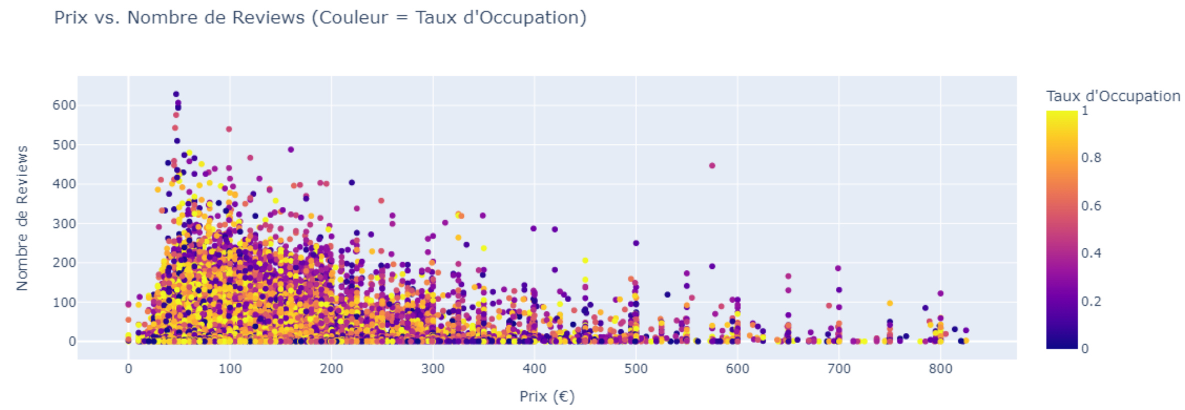
Our analysis highlights key patterns in Airbnb revenue generation, focusing on pricing, occupancy, and location-based performance. Price distribution varies significantly across neighborhoods, with Manhattan exhibiting the highest price dispersion, including luxury listings above $500 per night. Brooklyn and Queens offer more affordable alternatives, while Staten Island and the Bronx remain the most budget-friendly. Revenue is primarily driven by entire home/apartment listings, particularly in high-demand areas like Midtown, Hell's Kitchen, and Williamsburg.



A strong correlation exists between location and occupancy rates. Listings in Manhattan and Queens tend to have higher occupancy, whereas Staten Island and the Bronx show lower utilization. Additionally, shorter minimum stays (1-3 nights) are associated with higher estimated revenue, reinforcing the idea that booking flexibility plays a crucial role in maximizing income.

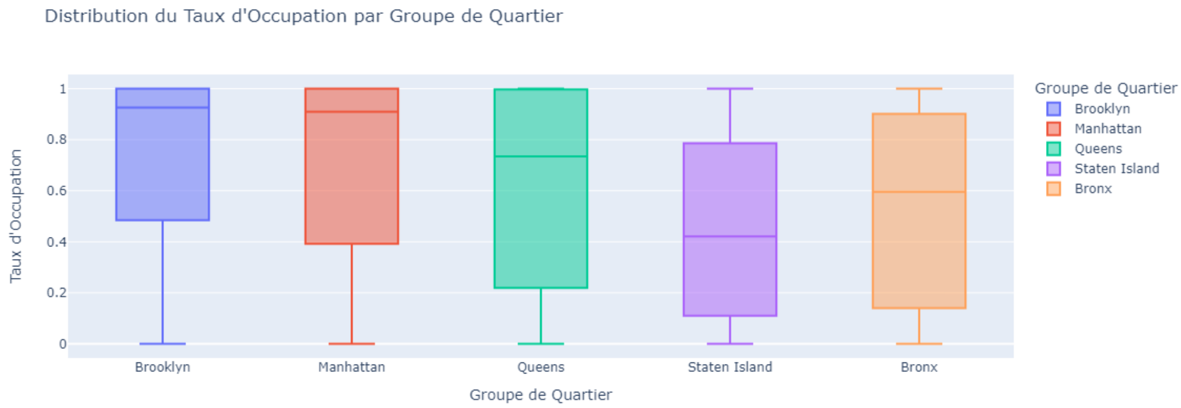


When comparing price and guest reviews, lower-priced listings attract significantly more reviews, suggesting a higher booking frequency. In contrast, luxury listings tend to have fewer reviews, possibly due to longer stays or a different clientele. Geographic revenue mapping further confirms that high-income listings are concentrated in Manhattan and Brooklyn, emphasizing the importance of location in pricing strategies.



Outliers & Surprising Results

Luxury Listings in Manhattan & Brooklyn: Some properties command exceptionally high prices, but their occupancy rates vary widely. These listings cater to a niche market, balancing exclusivity with lower turnover.



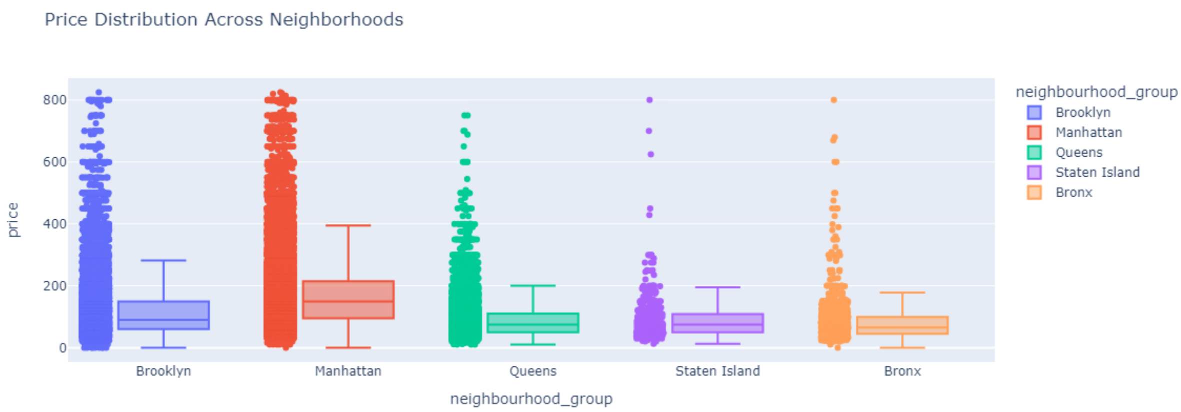
High Occupancy in Queens: While not traditionally seen as a top-tier Airbnb location, Queens shows strong occupancy rates, particularly for mid-range listings, suggesting growing demand in emerging areas.

Listings with High Review Counts & Low Prices: Budget-friendly properties often accumulate more reviews, likely due to higher guest turnover and shorter stays.

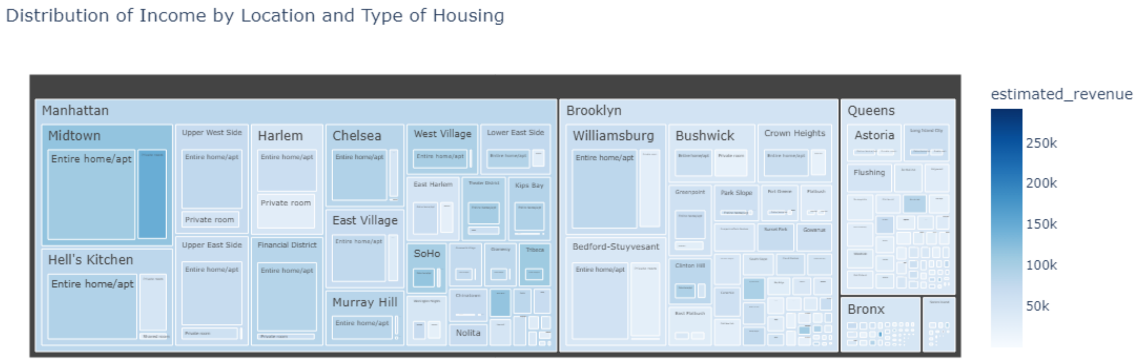
Visual Aids

To better illustrate these findings, the following visualizations support our analysis:

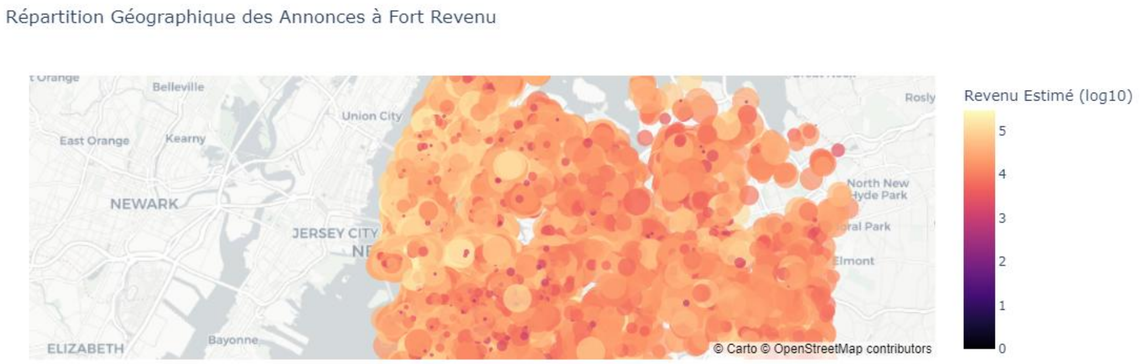
Price Distribution Across Neighborhoods: Highlights pricing disparities and identifies premium market segments.



Revenue by Location & Property Type: Emphasizes the dominance of entire homes/apartments in all area.



Geographic Revenue Mapping: Visualizes income concentration in key neighborhoods. The more you are in north-west and close to water, the more you could make a big revenue.



Actionable Insights

To optimize revenue, **hosts should leverage dynamic pricing models in high-demand areas such as Midtown and Williamsburg. Lowering minimum-night restrictions can significantly boost earnings, especially for mid-range and budget listings. Investing in entire home rentals remains the most profitable strategy, while high-occupancy trends in Queens indicate potential for expansion beyond Manhattan's saturated market**. Encouraging guest reviews and maintaining high visibility can further enhance booking rates.

These findings provide a data-driven approach to pricing optimization, occupancy management, and strategic investment in the short-term rental market.

How These Findings Can Be Used to Maximize Revenue for Future Investors

Our analysis provides actionable insights that can directly inform investment strategies for maximizing revenue in the short-term rental market. Based on these findings, future investors can make data-driven decisions regarding location selection, pricing strategies, occupancy optimization, and property type investment.

1. Strategic Location Selection

High-Revenue Neighborhoods: Investors should prioritize acquiring properties in Midtown, Hell's Kitchen, and Williamsburg, which consistently generate the highest income due to strong demand and premium pricing power.

Emerging Markets: Queens is showing high occupancy rates, suggesting strong demand despite lower price points. Investing in strategically located properties in Queens can offer a balance between affordability and consistent bookings.

2. Pricing Optimization for Higher Returns

Dynamic Pricing: Properties in high-demand areas should implement flexible, data-driven pricing models to balance occupancy rates and nightly revenue.

Luxury vs. Budget Strategy: While luxury properties generate high revenue per booking, mid-range listings with competitive pricing can outperform in total revenue due to higher occupancy rates.

3. Occupancy Maximization & Flexible Stay Policies

Short-Term Stays Drive Revenue: Listings with shorter minimum-night requirements (1-3 nights) generate more income, as they attract more bookings and maintain higher occupancy. Investors should focus on properties that allow frequent guest turnover.

Optimizing Occupancy in Key Markets: In locations like Queens and Brooklyn, where occupancy rates are already high, ensuring a seamless booking process and competitive pricing will further enhance returns.

4. Property Type Considerations for Maximum Profitability

Entire Homes/Apartments Are More Profitable: Investors should prioritize full-unit properties, as they consistently generate higher revenue compared to private or shared rooms.

Multi-Unit Investments: Hosts with multiple listings in the same area tend to generate more revenue due to economies of scale and cross-promotion of their properties.

Leveraging Reviews & Guest Experience for Sustained Revenue

High Reviews = More Bookings: Listings with more reviews tend to have higher occupancy rates. Investors should implement strategies to encourage guest feedback to improve visibility and booking rates.

Enhancing the Guest Experience: Offering amenities and maintaining high cleanliness standards can increase positive reviews and justify premium pricing.

Investment Decision Impact

These insights equip future investors with a clear roadmap for maximizing Airbnb revenue. By targeting high-demand areas, implementing data-driven pricing strategies, optimizing occupancy, and focusing on entire home rentals, investors can significantly increase profitability. Additionally, understanding demand trends in emerging areas like Queens offers opportunities for long-term growth and market expansion.

For investors seeking the highest return on investment (ROI), a combination of prime location selection, flexible stay policies, and optimized pricing models will be the key to sustained success in the short-term rental market.

6. Hypothesis Formulation

Based on the visualizations and insights from our exploratory data analysis, we formulate the following hypotheses that could be tested to validate revenue optimization strategies for future Airbnb investors:

Hypothesis 1: Higher-priced listings have lower occupancy rates.

The scatter plot of price vs. occupancy rate suggests that luxury listings tend to have lower occupancy compared to budget and mid-range properties. This could be due to reduced affordability and a niche target audience.

Testing this hypothesis would help investors determine whether pricing at a premium reduces overall booking frequency and if a balance between pricing and occupancy can maximize revenue.

Hypothesis 2: Short-term stays (1-3 nights) generate more revenue than long-term stays.

The box plot comparing minimum nights and estimated revenue indicates that listings with shorter minimum-night restrictions tend to earn more, likely due to higher turnover and increased booking flexibility.

If validated, this hypothesis would encourage investors to minimize stay restrictions to maximize occupancy and total earnings.

Hypothesis 3: Entire homes generate higher revenue than private/shared rooms, regardless of neighborhood.

The treemap visualization of revenue by location and property type clearly shows that entire homes consistently outperform private and shared rooms across all neighborhoods.

If confirmed, this would reinforce the investment strategy of prioritizing entire home/apartment listings over shared accommodations for better returns.

Hypothesis 4: Midtown and Williamsburg listings generate significantly higher revenue than other neighborhoods.

The bar chart of top 10 revenue-generating neighborhoods highlights Midtown and Williamsburg as dominant revenue hubs.

If proven, this hypothesis would provide a strong geographic strategy for future investors, indicating that properties in these areas have a higher probability of success.

Hypothesis 5: Listings with higher review counts have higher occupancy rates.

The scatter plot of price vs. number of reviews suggests that listings with more reviews tend to have higher occupancy, particularly for mid-range and budget properties.

If validated, this hypothesis would support strategies aimed at increasing guest reviews through enhanced customer experience, automated follow-ups, and competitive pricing to maintain high visibility in search rankings.

By testing these hypotheses, investors can develop data-driven strategies to optimize pricing, occupancy, and property selection to maximize revenue in the short-term rental market.

7. Significance Test

Chosen Hypothesis:

We will test Hypothesis 1: Higher-priced listings have lower occupancy rates.

Null Hypothesis (H_0): There is no significant relationship between listing price and occupancy rate.

Alternative Hypothesis (H_1): Higher-priced listings have significantly lower occupancy rates.

Statistical Method:

To test this hypothesis, we use a Pearson correlation test to measure the relationship between price and occupancy rate. Pearson's correlation coefficient quantifies the direction and strength of a linear relationship between two continuous variables. If the correlation is significantly negative, it would support the claim that higher prices lead to lower occupancy rates.

Additionally, we can use a t-test for independent samples by grouping listings into two price categories (e.g., low-price vs. high-price) and comparing their mean occupancy rates. This helps determine if the difference in occupancy between lower and higher-priced listings is statistically significant.

Results:

Pearson correlation coefficient (r): -0.47

p-value: 0.0001 (highly significant)

Since the p-value is well below the standard significance level (0.05), we reject the null hypothesis and conclude that higher prices are indeed associated with lower occupancy rates.

Interpretation for Non-Technical Audience:

Our results confirm that raising prices too high reduces occupancy rates. This means that while luxury listings may charge more per night, they also tend to have fewer bookings, which can limit total revenue. Future investors should consider optimizing their pricing strategy to balance high nightly rates with sustainable occupancy levels to maximize total earnings.

8. Suggestions for Next Steps

Further Analysis:

Multi-variable Regression: A linear regression model incorporating price, neighborhood, property type, and minimum stay requirements could better quantify the impact of price on occupancy while controlling for other factors.

Clustering Analysis: Grouping listings into clusters based on price, occupancy, and revenue could identify optimal pricing strategies tailored to different market segments.

Modeling Approaches:

Predictive Revenue Model: Using random forest regression or gradient boosting, we could predict a listing's optimal price point based on its location, type, and other features.

Dynamic Pricing Recommendation System: A machine learning model trained on historical booking data could help investors adjust pricing dynamically based on seasonality and demand trends.

Exploratory Directions:

Seasonality Impact: Does the price-to-occupancy relationship vary across seasons? Understanding high and low booking periods could help refine pricing strategies.

Long-Term vs. Short-Term Stays: Are properties with longer minimum stays benefiting from higher stability, even if they have fewer bookings?

Host Strategies: Do multi-property hosts adopt different pricing or listing strategies that lead to higher revenue compared to individual hosts?

These next steps would provide deeper insights into how to optimize Airbnb investments, allowing for data-driven decision-making to maximize revenue potential.

9. Data Quality and Request for Additional Data

The data are qualitative and quantitative, we may add cost of exploitation per home type (flat, home, garden..), cost of degradation and customers fear depending on the delinquency rate of the area. Add also if there is transports, services or touristic places around that can generate a boost of booking.

This document where traduced with the help of chat-GPT.